Nokia Core Node BLR-CORE-02 QoS Congestion Event SOP

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Warning | **Alarm Code:** ALARM_3127

Description

This SOP outlines the steps to troubleshoot and resolve a QoS congestion event on Nokia Core Node BLR-CORE-02. QoS congestion events occur when the node's QoS policy is unable to handle the incoming traffic, leading to microbursts, interface tail drops, and QoS queue starvation. On Nokia Core nodes, this is typically due to misconfigured QoS policies, oversubscription of interfaces, or inadequate buffer sizes. The node's hardware and software components, such as the Route Processor (RP) and Line Cards (LCs), play a crucial role in managing traffic and enforcing QoS policies. When the node's QoS policy is unable to handle the incoming traffic, it can lead to a range of issues, including packet loss, jitter, and latency. This SOP will guide the troubleshooter through the diagnostic and resolution process, ensuring that the node is restored to a stable state.

Pre-requisite Checks

- Pre-requisite: Verify that the site is accessible and the node is reachable via the CLI or SSH.

- Pre-requisite: Check the node's redundant link status to ensure that it is functioning correctly.

- Pre-requisite: Review the node's configuration to ensure that the QoS policy is correctly configured and applied.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy	High
Oversubscription of interfaces	Medium
Inadequate buffer sizes	Low

Diagnostic Commands & Expected Outputs

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Command: show qos policy, Expected Output: Policy name, Policy type, Priority, Description: This command displays a list of configured QoS policies, including their names, types, and priorities. It helps determine if the QoS policy is correctly configured and applied.

Command: show interface, Expected Output: Interface name, Status, MTU, Description: This command displays a list of interfaces on the node, including their status and MTU. It helps identify if any interfaces are oversubscribed or experiencing issues.

Command: show buffer, Expected Output: Buffer size, Buffer utilization, Description: This command displays a list of buffer sizes and utilization on the node. It helps determine if the buffer sizes
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are adequate to handle incoming traffic.
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Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy, Description: Review the output of the show qos policy command to ensure that the QoS policy is correctly configured and applied.

Step 2: Step: Adjust QoS policy priority, Command: configure terminal; qos policy priority , Description: If the QoS policy priority is found to be incorrect, adjust it to the recommended value using the configure terminal command. This ensures that the node enforces the correct QoS policy.

Step 3: Step: Adjust interface MTU, Command: configure terminal; interface mtu , Description: If an interface is found to be oversubscribed, adjust its MTU to the recommended value using the configure terminal command. This ensures that the node can handle incoming traffic correctly.

Step 4: Step: Adjust buffer size, Command: configure terminal; buffer size , Description: If the buffer size is found to be inadequate, adjust it to the recommended value using the configure terminal command. This ensures that the node can handle incoming traffic correctly.

Post-Check Verification

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Command: show qos policy, Expected Output: Policy name, Policy type, Priority, Description: Verify that the QoS policy is correctly configured and applied by reviewing the output of the show qos policy command.

Command: show interface, Expected Output: Interface name, Status, MTU, Description: Verify that the interface MTU is adjusted to the recommended value by reviewing the output of the show interface command.

Command: show buffer, Expected Output: Buffer size, Buffer utilization, Description: Verify that the buffer size is adjusted to the recommended value by reviewing the output of the show buffer command.
```

Roll-back Procedures

- Step: Revert QoS policy priority, Command: configure terminal; no qos policy priority , Description: If the fix fails, revert the QoS policy priority to the previous value using the configure terminal command. This ensures that the node returns to its previous state.

- Step: Revert interface MTU, Command: configure terminal; no interface mtu , Description: If the fix fails, revert the interface MTU to the previous value using the configure terminal command. This ensures that the node returns to its previous state.

- Step: Revert buffer size, Command: configure terminal; no buffer size , Description: If the fix fails, revert the buffer size to the previous value using the configure terminal command. This ensures that the node returns to its previous state.